



River deltas in crisis

All around the world, the fertile river deltas and the millions of people living there are facing an existential crisis. <https://digit.in/aug20-15>



Evidence for hotspots

This paper classifies climate change hotspots into three types, deltas, semiarid regions and glaciers. <https://digit.in/aug20-16>

the water levels in the low lying cities below the tropics, shrinking fertile land and causing more fresh water to end up in the salty ocean. About 40 percent of the world's population lives within 100 kilometers of a sea coast. The disappearing ice also means that the planet reflects back less sunlight, trapping even more heat. These are the many connected aspects of the global climate that are all falling like a series of dominos. The wealthy nations of the world have better infrastructure to deal with the problems that emerged, the ones that are most severely affected will be in the developing countries.

PROBLEMS IN INDIA

Because of the many river deltas in the deccan, India is particularly vulnerable to climate change. The patterns of the monsoons are becoming increasingly more erratic, leading to situations where farmers cannot reliably grow crops. In India, this can lead to periods of drought followed suddenly by flash floods which washes away the crops. This is what happened in 2019, where the rainfall was unexpectedly sporadic early in the monsoon region. Later in the year, there were flash floods in 11 states leading to the loss of 1,200 lives. The problem is exacerbated by rising temperatures melting off more glaciers in the Himalayas, which causes the water to come rushing down the rivers. Added to that is the mismanagement of the dams. Back in 2011 itself, a commission headed by Madhav Gadgil had presented a report to the government where they had clearly outlined the possibility of floods if there was mismanagement in the dams. The warning went unheeded and in the Kerala floods of 2018, 483 people died. The report recommended planning the release of dam water in periodic intervals, but the waters were released whenever the dams got full, without much coordination. The Maharashtra Government has established a task force specifically to address the new problems caused by climate change, and the central government is formulating new laws for the management of river basins. Wide-

spread mining and construction in the western ghats also contributed to the landslides. Around Mumbai, unplanned development means that the residents of housing colonies in satellite cities often find themselves in flooded homes, with destruction outside, cut off from supplies.

When there is widespread industrial activity and mining in the delta regions, the surface tends to become weak and sinks. This causes the ground level to dip, increasing the damage caused by floods. India remains one of the few countries in the world where the number of deaths caused by flooding continues to increase over the years. The shift in monsoon patterns

atmosphere. Within the decade, we will also have to find ways of scrubbing the atmosphere of the carbon that has already been dumped into it.

One of the simplest approaches is to help the planet clean itself, because the world is equipped with natural carbon scrubbers and these are known as forests. Unfortunately, agricultural expansion is reducing forest cover around the world. A move towards more sustainable agriculture is important, where a combination of genetics, fertilisers, accurate forecasts and water management practices can be used to increase the yield on the available land.

There are, however, artificial ways to remove carbon dioxide from the atmosphere. The approach of increasing energy efficiency as well as reducing emissions are together called carbon sequestration. The move from petrol or diesel based vehicles to electric ones is something that will help in a big way. Efficient public transportation systems such as Hyperloop also hold a lot of promise. Autonomous vehicles promise not only to be more



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People struggling on a rock during floods in Jharkhand, 2019

is not clearly understood yet and more studies need to be done to understand what the new patterns are.

All around the world, river deltas make up for only 1 percent of the surface, but considering how fertile they are, these regions are occupied by half a billion people. These will be among the most hit because of climate change. To mitigate the worst effects of climate change, it is necessary to invest heavily in these regions first. This is where technology and science can come to the rescue. As far as India is concerned, planned development, better weather forecasting, and coordinated dam management can go a long way.

POSSIBLE SOLUTIONS

The most important thing that the people of the planet must find a way to do is to reduce the indiscriminate dumping of greenhouse gases in the

efficient, but the coordinated movement of fleets for goods transport over land also promise to reduce energy use considerably. Another idea is to capture the carbon at the source of emissions, from powerplants or industries, and bury them in place of mined coal seams and depleted oil reservoirs. The idea is to put the carbon back in the ground after using the fuel, instead of the atmosphere. The Sleipner A project in the North Sea near Norway is one of the few commercial carbon injection projects in the world where carbon is actually pumped back into a saltwater reservoir 800-1000 meters below the ocean. Better management practices and new better technologies both promise to reduce emissions and make energy use more efficient. Basically, the sooner the world switches from fossil fuel to greener alternatives, the better it is for the planet. 